

ActInf Livestream #050.2 ~ "Interoception as modeling, allostasis as control"

Livestream | 1:58:28

Thank you.

Hello and welcome. This is ActInf Livestream number 50.2. It's November 3rd, 2022, and it's our third discussion on this paper.

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All right. We are in livestream 50.2 in our third discussion on interoception as modeling allostasis as control.

And we had a .0 background and context video. And two weeks ago now, we had a great number 050.1 discussion.

So we're just going to have introductions of people who were here last week, and then Ian will share something for a few minutes.

And then we'll be discussing whatever people want.

So we can each just say hello and anything that we want to explore today or some reflection from the last two weeks.

So I'm Daniel. I'm a researcher in California. And I'll pass to Dean.

Thanks, Daniel. I'm Dean. I'm here in Calgary in Canada.

And not much to say other than I'm looking forward to getting past the content, which was the zero, the context, which was the one,

and figuring out what we can conceptualize into in this too. I'll pass it over to Jordan.

Sure. So I'm Jordan. I'm a postdoc working with Lisa Feldman-Barratt and Karen Quigley at Northeastern.

And I think I said the last time that we've done this, this paper sort of emerged out of the energetics reading group that we've done.

So I'm broadly interested in brain energy metabolism. I've been doing some other work on that.

And I was not here last week in the week between these sessions because I was at the International Conference on Brain Energy Metabolism, which was very fun.

And yeah, I think I'm excited to get into the conceptual details of this paper here too.

And interested in thinking about what a control theory model of the brain and of interoception looks like.

I'll pass it over to Eli maybe.

Hi, I'm Eli. I'm the first author on the paper, grad student with Hjelndr-Wilhelm Vandemeind, who is sort of our undermentioned co-author.

And with Lisa Feldman-Barratt and Karen Quigley.

I would say that I just got away from a conversation with Maxwell Ramstead.

So I'm actually a bit excited about Active Inference.

He really clarified for me what it is and what it means.

Yeah, and I think I could pass to...

Is that all of us?

Yep.

Or Ian, do you still need to go?

Hi. Yes, thank you.

So I'm Ian. I'm based in England, UK.

And so my background many years ago was in kind of undergraduate of biochemistry and then PhD in molecular biology applied to immunology and how white blood cells can determine between molecular signals of safety and threat.

I didn't stay very long with that.

And then I moved on to working on environmental management projects and therapy.

So manual therapy, health mentoring and related things.

And in that world, a common kind of theme I noticed was paying attention to bodily signals and sensing and interpreting bodily signals.

But I didn't have a way to talk about that confidently that made sense with the versions of homeostasis and the autonomic nervous system that I was taught in the 90s.

So when I, a few years ago, discovered there was a small group of researchers studying interoception, I got very excited very quickly.

And I've been involved recently in a little bit of research on training for improved interoceptive awareness with a university nearby here.

And when Daniel shared that this paper and said it was going to be on the active instruments institute, I got giddily excited and wasn't quite sure why that was.

So I said, when I was trying to figure it out and watching 55.0, I'm asked if I could share a few pictures, which maybe I'll do now if I can share screen.

To, oh, is that going to be possible?

Sure.

And I'll crop it so that it's the whole screen.

Okay.

So I thought, why was I so excited to see this paper?

And I hope it might be able to help people in the therapeutic world.

Because I know that you've sort of talked about physiology, meeting engineering.

But then this is a slightly different, maybe a slightly different take.

Yeah, go full screen.

All right, just give me one.

Oh, there we go.

Yeah, give me one second.

So, oh, is that, can you see that now?

Yep, I'm just going to make it big.

Just give me one second to get it finished.

All right, go for it.

So in the .0, I heard people talking about baked in reflexes.

And it made me think of these pictures, which have been going around on the internet of why, sorry, why are cats scared of cucumbers?

Is it because they're born to be scared of snakes?

So kind of some kind of genetic, you know, cucumbers look like snakes.

So they go from feeding mode into fear mode.

Or is it that their cruel owners have just put something surprising behind them?

Which, so it's just stimulating a startle reflex.

Either way, I don't know what the cat's interceptive experience is.

But there certainly seems to be something that is automatic is happening there.

So thinking about in the therapeutic world, there's what's often, what we're often trying to achieve is if something is, if a behavior has become maladaptive and starts to feel like it's reflexive for a person.

So let's say a phobia against spiders or a phobia against public speaking or a phobia, you know, a fear of going into public spaces, then the person may feel like, I know I'm going to suffer in this situation.

It's just a reflex reaction.

But with interoceptive exposure therapy, what I find is interesting is that people are able to, in a safe space, able to say, okay, when this thing happens to me, I feel maybe.

Short of breath, tightness in my chest.

So if we can recreate that sensation in a safe environment, can gradually they learn to change whatever their predictions or their beliefs are about what that sensation means.

And it seems to work.

And, you know, one method of doing that for, if breathing is an issue for people with anxiety is to recreate it by breathing through a straw.

So the, you know, the aim really is to turn something maladaptive into a more adaptive behavior or a vicious cycle into a virtuous circle.

So instead of triggering this kind of reaction, which feels unpleasant and is involved, you know, reflexes or supposed reflexes are being triggered that are linked to shortness of breath or increased heart rate.

Instead, they can learn to create some changes in the body that might be associated with something more pleasurable, such as, you know, relaxing with family on a beach or some kind of pleasant situation.

So different changes in their autonomic nervous system.

And in general, what we, you know, what we are sort of interested in doing is using inter interception as an access point to be able to untangle, you know, the story, the thoughts that people have from the situation they're in and the bodily sensations that they're sensing or that are arising.

So, Dean, you mentioned about packing your parachute.

So Dean, you mentioned about packing your parachute.

This was another kind of image that came to mind.

Most babies are probably born with some kind of reaction that stops them throwing themselves out of high places, but it does seem to be possible to override any interoceptive fear and for some people even to enjoy throwing themselves off high places.

Likewise, we can learn, you know, we can learn that situations such as maybe sitting in front of a laptop with an email box full of emails for some people can lead to an interoceptive sensation.

It's like a sinking feeling or a cramping or, you know, a heavy feeling.

So then they've learned, they weren't born with that, but they've learned to associate these situations with something really, you know, stressful.

Ah, this was from the paper, I think, Jordan or Eli, you referred to last year, sorry, last week, the dot one.

I haven't digested it fully, but it, you know, it's really becomes apparent that the direction of these circles and the loops within loops,

are quite complex, but I really liked the way that you described, these are my reflections now from what, where you ended in dot one, which was this idea of kind of cooperative control or optimal control.

And, you know, I was sort of taught about feedback loops as being like thermostats and, you know, maybe fat levels in a person's body being influenced just by the hormone leptin, for example.

But it seems that the set points in the settling ranges are much more mobile than once was thought.

And we see that now with, you know, and involve a lot more crosstalk.

So rather than just targeting something like leptin as the problem for the obesity sort of crisis, then, you know, this is just a stat taken from UK government office,

which created an obesity systems map that identified, you know, 108 interconnected variables that determine people's energy balance in relation to obesity.

And some of those things might include marketing and advertising, social pressures.

So how do we, you know, how do we bring all of those things in to these types of models that are being created?

And then there's epigenetic factors as well.

So, you know, people can inherit their exposome through epigenetic changes and can that change their intrareceptive predictions?

And, you know, how I think about this from a kind of health mentoring or coaching.

Well, if this, I'm not quite sure if I've wrapped my head around time averaged the concept of time averaging completely.

But, you know, if I have someone comes to me and they say, we're talking about the health and they want to be healthy enough in 20 years time to be able to support their grandchildren and play with their grandchildren,

then that's if that's a strong enough interceptive signal of pleasure into the future, you know, or strong enough desire, can that override the short term interceptive prediction of bliss,

bliss, which is maybe achieved from some, you know, very predictably produced junk food that's been designed to create a prediction of, you know, an interceptive anticipatory pleasurable sensation.

How do those over time, how do those two different kind of competing desires work with each other?

Is it a battle of will or what's, you know, some of them seem really deeply reflects, they both seem deeply reflexive, but they are often at odds with each other.

And again, with this kind of time averaging I was thinking about in the allotment, this isn't from my allotment, but you know, plants when they're stressed, they bolt early and flower early if they think their environment is dry or not,

haven't got enough nutrients or enough space for the roots, then they will flower early go to seed early.

So is that what time averaging is related to?

And the other thought was about this idea of someone mentioned about exercise being good for you, because it's a short term stressor, and it helps your heartbeat slower over the long term.

I see lots of sort of people talking about hormesis in the health and fitness and hormetic stressors in the health and fitness kind of arena, but we don't seem to have a good model of it or a way of telling what the sweet spot is for a hormetic stressor.

Yes, and there's lots of, you know, people doing hormetic stressors at the moment, Wim Hof and hot and cold therapy and so on.

And the last thing I'm nearly done, the last thing that interested me after was this idea of ergodicity, and the distinction you made in this paper between ergodic and non-ergodic sort of ergodic systems.

And I was at a parkrun last Saturday, and talking to some of the people, my friends at the finish, about how difficult it is to, you know, you can feel like you're,

you're interceptively feel like you're running really bad one day, but then your time is actually quicker than what you expect.

Another day you can be really up for it and excited and feel buzzing, and you get to the finish line and realise you've gone slower.

So this picture is from someone who broke the world record, she did a marathon in under two hours, and part of the tools in their toolbox was to have a laser to keep their speed constant throughout the marathon.

So, you know, the both at any point, either in one instance or throughout the whole marathon, they, their time was constant.

So I guess my question is, is technology, if we've, if we're trying to design systems based on the idea of ergodicity, will we, are we reverse engineering that into our lives, maybe?

And where would that lead? If you've seen the, the, the film, Don't Look Up, and it's all about prediction and modelling.

So what will we, what will we be evolving into if we don't have, you know, the right systems that are truly matching biological, what's actually happening with our biology?

And yeah, that was, they're my reflections.

That's great.

Awesome, Ian. Thanks for bringing in all of that. That's really insightful.

So I think we have more than enough to just begin. So let's just start with whoever wants to reflect, please. Jordan first.

I mean, yeah, I guess I can, this will be going a little bit off of the paper here, but I think just what you were bringing up is, is interesting to, um, some of the other work that I've been doing as well, too.

And what we can think of, what we can think of sort of extending the systems that we're talking about in this paper outside of the body, right?

So some of what you were bringing up was you were talking about, um, some of the sort of social feedback that you get for, um, maintaining allostasis.

Um, you were bringing up the example of email and the sort of interoceptive response that you can get from anticipating, um, stress from, from email, from communication, these sorts of things.

And you're talking about how there's, how external references, and you were giving the example of technology and a laser to keep pace on a marathon can help, um, can basically act as an external, you can use external reference signals.

Right.

And I think one way to think about social influence or some, some of the social forces on interoceptive experience is also as like external reference signals, right.

Or it's at least as external factors.

And so the, um, I think one thing, so you can think about the, the example of the email that you were giving is it's an interesting case, right?

Because you basically have, um, in some ways you're kind of opening yourself up to this import, to this portal of potential interoceptive disruption that can arrive at any moment.

And that's not going to be, um, it's not going to have some of the usual, uh, contingencies that would allow you to predict it.

Like you would in other cases.

Like if you're going to a meeting, you're going to your, you're going to go meet with your boss, you know, it's at four o'clock.

You know, that none of the sort of social signals from that, that could potentially be something you have to react to something that you have to adapt and then tear up the state to something.

Something that's going to cause you to have to leverage or basically mobilize your body to respond to that situation.

Those aren't going to happen until about four o'clock.

When you go to that meeting, if you're open to email, you're basically going to be able to be prompted within any sort of work hour.

So you need to be prepared for that potential mobilization at any point, right?

If your boss is like a real, you know, if your boss is really awful, then maybe you're going to get emails in the middle of the night.

And if that's the case, then you need to be prepared for some sort of, some sort of allostatic mobilization at literally any point of the day, which is going to bring with itself some sort of consequences of needing to have this system in a state of preparedness.

Because disruptions could come at any point, right?

And so I think the, if one of the principles that's in Eli's paper here, right, is that we're defining, putting together an allostatic response as being, keeping variables at multiple levels of this control system in a state where they can be maximized.

And so if you're in a state where they can be maximally responsive to the sorts of threats that could emerge, then that's going to look very different if you're constantly in a state where you need to be prepared for a lot of different disruptions that could come in from social sources.

Versus if you're in a state where you could rule some of those things out, you know, it's the weekend, you know, your boss or colleagues aren't going to call you.

And so you can not have to have some of these variables in a state of preparedness to immediately react to those sorts of things, right?

[illegible]

lieu of having to perform the calculations to regulate a lot of other stuff that's changing in your body over the entire period of it right it's and i'm thinking about it in terms of you do a similar thing with like a metronome or something like that as well right where if you're trying to coordinate a lot of different people to perform musical notes in time you'd want to have a conductor you'd want to have um you there's there's a real use in having external reference points for managing those things because literally everything in your body's changing as you're performing this stuff and commands need to be commands need to be given to to regulate in the face of all of that change eli's pointing so i think eli's got a point um yes i mean i was just going to say like we know that as you move a bunch of your sensory and interoceptive signals change so they're basically never actually constant in the first place if there's something you want to hold constant then yeah you need something like the laser that effectively gives you an artificial source of constancy a source of constant constancy outside the body because you don't have one internally

yes yeah and it's what what's also occurring to me um as you're saying that is as well as the internal conditions changing you know the the external conditions are changing and you know maybe getting windier at various points of the course um and there may be you know uh daniel you talked about um when we spoke a while ago about ants if they when is it appropriate for them to react to weather changes and there may be you know we want a marathon runner to essentially ignore all the weather changes um and just keep running at the desired speed so it's almost um yeah maybe that you know the last slide i put there was a bit of a a distraction really i don't think it's a distraction at all though but because i think that the no i think it's super super useful because like part of again this whole idea of um multiple variables in the body changing all at once that's kind of core to what we mean when we're saying allostasis right we're saying that there aren't any particular constants that are there there's a lot of variables that are moving and changing and what eli's making the point in this paper is that in lieu of set points in a lot of those cases what you want is some variables to try to occupy a zone of maximal flexibility to changing circumstances um

but at some point like it there might be some variables deep deep down in turn like very very central to regulating the body that are going to need to be kept at some close to homeostatic level and so one of the things that was coming up for me and working with the brain energy metabolism conference is that variables like blood oxygenation might be one of those right those are very tightly regulated if that strays outside of some very narrow parameters you're going to have really serious problems but if you think about your runner running the marathon the um that blood oxygenation level is going to be fluctuating it's going to be challenged there's going to have to be responses there are other compromises made in the body to maintain that and if that's the very if that's one variable that's being kept at a tight that's being tightly regulated at the core of interoceptive information other compromises that are going to happen to regulate that down the line might be changes in muscle force delivered changes in speed a lot of fluctuations that are happening in terms of how they're moving their body through this race and so if you're giving them some external external sensory signal to couple to basically at the other end of this control hierarchy that eli's describing

right then you're giving them something at the other end of it to couple to to make their movement through the environment a bit more regulated and predictable so you can think of it as the entire control hierarchy that eli is describing in this paper we have some very closely regulated interoceptive values far to the left we have a deep net of predictions and prediction error throughout the entire cortex that's performing some regulation um across the entire network and then we have exteroceptive information at the other end of it and what you're doing with the lasers you're basically giving the organism the goal of fixing some of those values at the exteroceptive layer to give some more structure as they move throughout this entire deep deep deep network in all the space between right yes and what i'm thinking also that you know what the marathon runner did before that marathon so the marathon was made predictive his work his breaking that record was made easier for him by the laser but also you know maybe he influenced his blood his blood oxygenation you know this the upper and lower limits of the acceptable settling range by doing um altitude training to stimulate more um red blood cells he might have done hyperbaric therapy um to all those things which will make it easier for him just to follow that later right and i guess in eli's framework right that would be that's not necessarily changing the settling point of blood oxygenation but what it is doing is at variables that are just slightly levels down from that you're eli would this be right to say you're basically widening out that sigmoid function so that there's more adaptability um from that additional training for increasing red blood cells that sort of thing is that right yeah definitely i'd also sort of add that some physiological variables are very tightly regulated but one of the open questions is is sort of how at how many removes are they from a disturbance so like blood oxygenation or the ph in the brain that you mentioned you know what does it actually take to disturb those right like blood oxygenation you can probably disturb by just having greater or lesser uptake of oxygen from the blood brain ph i'd imagine that honestly i'd imagine that you have to disturb a lot of things sort of disturb your way through the outer control system like the outer several layers of control systems before you can actually perturb that but you can think about things like the fact that you have a blood brain barrier in the first place is one thing that might help keep that sort of thing tightly regulated right exactly that's what i mean is that yeah you know living systems tend to be sort of homeostatic or allostatic at every level like a cell maintains its own internal you know homeostasis in some sense or allostasis by interchange with its environment then an organ system may also have you know such as the brain may have the blood brain barrier you know and sort of other levels of protection that insulated from its environment right because it has to be in cooperation with other organ systems for sure and i guess the thing that i'm saying is just that the for some of those parameters that do need to be really tightly regulated like that right you might see some some uh some naturally developing physiological structures like the blood brain barrier that are there to help keep those values tightly regulated as opposed to having it be more under behavioral control in which case there's some there's some flexibility there's some allostatic flexibility in those right the what you want is if there's something that really really really has to be kept at a certain level you would ideally have some of the evolved biology work to fix some of those parameters in the physiology

itself maybe what do you think i'm sorry actually could i just take my dog out for five minutes
i was thinking earlier oh sorry no go ahead go ahead jordan i was just going to joke about i want to go
find a cucumber and try that out of my cat too yeah actually um terry in the chat wrote irish cats don't do
this there are no snakes in ireland seriously i've tried it with our cat oh so not all cats well and
our cats lived indoors for her entire life except for the first six weeks or so which was probably
blind for half of that so i should try it out on her too and see if there was some learning component to
it or if she if she has it innate sorry dude yeah go ahead dean i just wanted to kind of throw this in
here and and i don't know i'm feeling kind of bad because eli's walking his dog but i'm sure you'll
you'll pick up the thread here so in the in the dot one i brought up this idea and i'm not sure how often
it's it's not ubiquitous yet but i i think that the focus in this paper is about perspective swaps so
what do i mean by that i kind of wanted to do a quick lap around that ergodically and then see
what you guys think about this so uploading to a laser as as a perspective uh in the dot zero daniel saying
staying within the connection space as he was talking about cdf to pdf as being another type
of perspective taken um the idea of bringing psychologists and their perspectives and
engineers and their perspectives together in in an institute or a lab setting as a as a different
type of that different perspectives colliding or at least paralleling um yeah yeah dance dodgeball
opposing teams each having a different perspective because they're wearing a different pinny
colored pinny um instrumental one of the things we talk about a lot in active inference is
instrumentalism and inaction or the idea of um sort of the embodied space versus the theoretical the
map yeah perspective and the territory perspective um and in this paper's case the specific feedback control
focus as a perspective right what i'd like to do is take that idea that there's all these perspectives
that one can be swapped in and swapped out within the idea of regulation within ranges yes but i'd like
to talk a little bit about what ian brought up which is i think is a really critical point and maybe
quite complimentary not corollary but complementary to what the paper was talking about you said if we do
something in a safe space now to me i'm going to ask you this to me safe sometimes means controlled
meaning the agents set the condition or at least they perceive they have some control over that
condition okay so that's a perspective that's sometimes perceived as ergodic because it's predictable
right then we have a sense of how do we strategize from that set condition one that's variability reduced
to one that's condition set or variability retained and i use the example that you had on one of your
slides which is 108 interconnected variables that determine dot dot dot because to me that's what this
paper is trying to focus on that specific feedback control is that shift in perspective that swap
between the variability reduced because that's what that's why i said the variability retained for me
as a skier was i've got an open field it's pristine it's 12 12 inches deep and away i go but as
more trees start amassing my ability to be free because of variability my my variability factor changes
and now the condition set is my pers my perspective swap has to be can i continue or am i just going to
end up bashing into a bunch of limbs right right so the bringing of the two together to me i want to kind
of make clear what i mean by perspective swap yeah i don't think it contradicts what you've written in
the paper i think the move from cdf to pdf is a perfect example of that perspective swap and i'm not

sure it's just a derivative because as i said in the last live stream you bring you bring psychologists and engineers into the same space and one doesn't get to rule over the other you need both to create that that swapping condition right right so right is that it is that essentially what specific feedback control helps us get a better sense of not just a grip or an optimality but sort of a a clear separation so that we can do the oscillating so that we can do the swapping and so that when we're having these kind of interactions we're not doing a transdisciplinary thing here we're really trying to get our heads around why did the laser work right because we can perspective swap that we might get closer to something satisfactory so i'm trying to understand so when you're talking about a perceptive or perception swap i don't know if i'm going to answer your question satisfactorily we might have to wait for you but i'm going to try my best here because i'm not the that's all i want you know because i'm trying my best too that's nobody's got to be perfect right right right and i think but i think that that's part of what factors into what i was going to how i was going to try to answer here which is that i think that um so

so the way i'll try to answer this is i'll try to draw on some of the social some of the social elastasis stuff that i've done which is involving more like you're saying interactions between people if we're bringing psychologists and engineers together this sort of thing so if let's say just if i answered your question right now let's let's pretend that i answered your question in a way that precisely got to the core of it and just stopped the conversation and you just said oh yeah i got it that answers it in a word right right then i might have perceptive i might have perception swapped slope so closely to the way that you would frame this question and so perfectly understood that the way that you'd asked it right that we don't actually explore the space around it in a way for other people to come in to understand anything of what's going on exactly if i actually didn't totally understand your question and i need to ask questions and we need to bounce this back and forth for like several minutes then we might still come to a settling point and we might perform some gradient descent to come to an actual answer here but because it's going to take longer it might end up bouncing around through different parts of the space of answers that we could look at and end up capturing other ways that um other perspectives might be able to understand it right so that they can catch some element of it that was predictable or that fit with some internal model that they'd already generated and then to be able to follow along to the gradient descent to reach down to that point right so i think that part of the advantage right of having interdisciplinary conversations and we've done this a lot especially you know in working with eli and then working with the engineers at northeastern these psychology engineer interdisciplinary collaborations right is that there will be a really long and frustrating period where people just do not understand each other um and you need to actually work through that because that's a period where you can do some learning and you can actually get some understanding of other perspectives other ways to build a model of the environment other ways to sort of take a perspective on events that are actually happening but um there will be a temptation through the entire thing to cut your losses and leave it

um because it is frustrating and it is metabolically costly i'd say um to do all of that information processing and to follow that process and it's not always obvious when the gradient descent is going to end either or when you do understand each other or when everyone is sort of followed into the loop and are able to follow that down to make the environment predictable to come to a shared understanding um you know it could have you might know when you're getting close but until you get to that point you might be in a really long and frustrating process of of search prediction error uh of uh basically bouncing around through the environment of trying to to come up with some model that can predict it right yeah so i feel like that was a very meta way of trying to answer your question but i think does that help i think there's a i think it gets it part of the use of of interdisciplinary work right and and how we uh how having these different perspectives are going to help us explore the space more fully because if we if we just had an internal language to each other if we perfectly understood each other in the first place then like you don't actually end up exploring the space in that case because you descend to you descend to sort of common predictions too fast to to really bounce around and explore everything you just you've basically described the capitulation to a retroactive or an abductive logical process that's all you've done and i don't think there's anything too nasty about that it's been around for over 100 years and and i think it's having a renaissance so thank you cool cool cool cool oh ian

oh yes as you were talking there about this you know um different disciplines coming together it was reminding me of um one of the active instruments institute's uh papers that were covered a few weeks ago on uh social communication as control i think it was um and also reminding me of you know the difficulty of a newborn child and a parent um you know that all the child can do is cry when it's feels some sort of discomfort and then the parent has to try and figure out what that that cry means does it mean it's hungry does it mean it's um nappy needs changing does it mean it's too hot or cold so there's a there's a period of learning each other's language and learning what each other's cues are and regulating each other and controlling each other um yeah that's what arguably that's all we're doing all the time as socially as adults we just got a more bigger set of vocabulary and um we can influence our environment more than the baby well then we can tell each other of what our what our expectations are too right like the baby is giving a really diffuse signal that there is some interoceptive state that's disrupted and then the adult has to like we were saying right explore around and bounce around until they can converge on that and get the baby satisfied to satisfy that interoceptive state right but you know if i

i so some of the other work that i've done is looking at how i really want to look at this paper that of uh of social communication is control because that's related to some other stuff that i've been doing too but is that um i think that you can think of social influence as um basically people regulating uncertainty in a social environment by um conforming to predictions that other agents have in it so you can think of a social norm as a set of collective expectations that people might have in an environment um and if you conform to those expectations then you can

minimize prediction error for people and you can minimize the likelihood that their behavior is going to change that your environment is going to become disrupted more entropic anything like that and so the baby's basically creating uh entropy in their environment for the adults in the room to try to minimize by satisfying interoceptive needs for them you can as an adult once you can communicate maybe get around some of those problems by saying you know you ask someone to pass the salt at thanksgiving or pass you know pass the butter you've made an expectation extremely clear and then people can satisfy that expectation and everything continues on predictably right they could also refuse and say no i'm not going to pass this on but if you do that you will move things onto a more unpredictable trajectory right you're going to uh push things out of the sort of bounds that people are used to navigating and you're going to potentially move off into this space of having to explore each other's reactions like like dean was talking about right yes yeah and what you're sort of saying there it's reminding me of um some of the work so it's a greek research researcher called dimitris xylyd gatas i think he's he's done a lot of work on ritual and um how rituals basically you know he's put heart rate monitors on people and measured cortisol levels and so on over you know the weeks and days following rituals and showing that it's you know there's a lot of synchronization that occurs during rituals um but the so the synchronization of physiology but that's bound up in these kind of codes or rules about how our plan or our group or our society should operate so it's kind of i you know my interpretation on that is it's training a person's physiology to align itself with social behaviors is this wait was this demetrius bolas demetrius um his surname is spelt x y g a l a t a s he wrote a book recently called ritual how seemingly senseless acts make life worth living but um yeah he's done some um study with measuring you know heart rate variability and stuff for fire walking kind of rituals and so on that's really cool jordan you did mention demetrius bolas and the work from 2020 through others we become ourselves the dialectics of predictive coding active inference also hinging on a shared first name but a very relevance paper as well um and then one thought to connect this idea of synchrony uncertainty controlled novelty in communication and in rituals that's when people are the laser for each other with their attention so then it's like okay we're gonna do a role play where this is gonna be the case i'll synchronize this perspective you'll have this perspective we'll just be like it will be just two stubborn characters having this argument it'll be very funny and then that is a predictable play it could even be scripted or at the very least you can have fun and so on so it's very interesting that that is reflected in the social space whereas the marathoning and going for the record which few people want to do most people can take a much more relaxed approach than the person trying to set the world record but in the social space uh as dean brought up there's a lot that we can bring to bear on the social space

from variability reduced settings like laboratory studies on cognition fmri studies with social stimuli but the social space is a variability retained area of application right and so there's a challenge with the translation yes jordan but but social norms right or sort of some social standards or mutual expectations add some variability constraint to that social space right because there's nothing physically stopping you from doing a lot of different things in any social setting but the point is that if you want to stay

in a variability reduced space and i think that there's some metabolic advantages and interoceptive advantages to staying in a metabolically or in a variability reduced state right then it's just the predictions of other agents that um set some of the bounds of where you'll wander into a variability increased state if you wander outside of those expectations or a variability reduced state if you stay within them right so as long as you're adhering to some predictions that other agents would have in that environment then you're going to stay in a variability reduced state and you can stay you can stay for lots of purposes sort of socially invisible to other agents you don't need to create new disruptions to an environment that you're going to need to be prepared to adapt to right

thank you dean sorry i'll lower my hand again so the fascinating thing about the paper was um i used a basic heuristic that says when in doubt and we're pretty much always in doubt zoom in zoom out and so when i when i look at the scale of what that would mean from the perspective of interoception those time scales are much tighter than if i zoom it out to the allostasis involved in running a marathon my zoom in zoom out range and potential gain is much different for the allostasis than it is for the interoception that's involved over that same course of time and so that's where i found the specific feedback control which implies that there must be a general feedback control as well so interesting that's where i realized part about the four different time segmentation possibilities allows for that zoom in zoom out um is that am i contradicting what the point was that you were trying to make in the paper eli um i'm i am slightly wondering where you got four specific scales well you have the I4 in terms of oh yeah oh that was just for modeling purposes no i understand that but i mean you can still you can still inflate that to yeah yeah but i think that's a pretty standard part of the hierarchical active inference literature isn't it yeah okay but do you think the idea that again we can swap from from the specific to the general is again it just seems to be again and again it's that ability to move within ranges but but the range doesn't necessarily remain consistent unless of course you're following a laser but that goes back to my original question is because we offload the the metabolic commitment that we would have to stay at that level of consistency we can then turn those resources over to other aspects of that endeavor

let me try to ask that a different way to jordan i think would be um would be how would different patterns in this type of a multi-timescale framing reflect different metabolic or energetic costs like does the brain use more oxygen glucose ketone bodies micronutrients who knows during what kind of activity such that there's what kind of a link between the physiological costs in gold yeah versus i mean cognitive angle i mean i think the way to to do it and this was i think um i was thinking of this when you asked gila too is that the the trick is right is that if you're regulating um some of the input to the interoceptive input that you're trying to regulate that might be really central to allostatic regulation right could be from sensed interoceptive variables uh like you know like blood oxygenation like blood glucose but the trick is right is that and this is that i think more what i'm interested in what i'm trying to develop with thinking about brain metabolism is that all of the steps and all of the communication that has to be done at some of these lower orders of this hierarchical scale right isn't free it has metabolic costs itself right and so

you know in the human brain the actual resting state cost of the human brain when it's just so you're just sitting still in the mri scanner is about 20 of your body's metabolic budget um it fluctuates a little bit as you're doing some sort of uh as you're doing sensory processing as you're dealing with unpredictable information the details of how it fluctuates like those matter too and we can get into those um there's relative changes in how much glucose you're consuming versus whether you're consuming the glucose with oxygen for like a more efficient yield of uh of atp energy but but the point is just that neuronal signaling itself is super expensive and if you're navigating an unpredictable environment then you can expect to have some of those expenses kick up or have them change in ways that are going to have to be regulated so what that means right is that even though there's a hierarchy of processing here where you're taking in sensory signals you're passing them up a cortical hierarchy and you're compressing them if you're doing that more and more and if you're in an environment where you have to be passing potentially more and more prediction error up this chain that can itself then be changing um important interoceptive sensed variables that might enter at different points of the hierarchy because there's still sense information but they might be more centrally regulated they might have downstream effects if those start to go out of whack right so basically i think that it like you brought up too that you know thinking about this from an allostatic perspective instead of a computational perspective and i think the computer metaphor that people have been using for the brain kind of computer scientists don't often think about cost in lots of ways too unless you know but they'll think maybe about time cost of how many cycles this like deep neural net is going to take to train but they often don't think about the fact that like all of the components of this computer do take like real energy to run and you know you can heat up the planet mining bitcoin you can do all of this stuff too there's external there's uh uh external costs to all of this that aren't taken into account when you're actually training these systems the brain can't ignore that because it is an actual biological system and its whole point is that it's going to have to keep that system working navigate the environment and grab metabolic resources to keep the cycle going and so um all of this computation that you're doing yeah isn't free it adds up and then can uh potentially have downstream effects on on sensed metabolic variables that are critical great thank you ian

uh thank you yeah i was just thinking then after um what jordan was saying about you know climate change and um that if that's a prediction that's maybe you know two decades ago wasn't going to affect my life but it might affect the lives of my children or my children's children to act now with that that's a that's a very abstract you know there's probably for most people there's nothing in their environment that would um give them that prediction of future threats in 50 100 years time so they have to create a very you know abstract sort of image of what the planet's going to be like and then to you know all the huge energy expenditure that's required to switch to renewables to you know organize society differently that's going to require a massive you know allostatic load in the short term or interceptive sort of yeah demand chronic low-level interceptive disturbance to create that shift through generations right and so i think you see right like there's if what your brain is regulating is

at the core of it some of the interoceptive parameters that are necessary to keep your body and your whole organism moving forward right then being thrown into a constant state of uncertainty is going to be costly if we think it's true that processing all of that uncertainty is going to have a metabolic cost attached to it so there's going to be some i think the way that that your sort of brain as a regulator is going to work is probably going to want to return you to some relatively predictable state and so the issue right is that if you're if you require if you can if you can model and understand externally right like if you can understand that in a long that the long term trajectory that you're on in terms of climate change is going to lead to disaster right it can still be really difficult to divert from that because the trajectory is at least is a predictable trajectory that you're on until it hits some of these critical exponential points where it's suddenly going to break right and so i think you can see that with climate change i think you can see that with the pandemic you can see a lot of cases where there's um there's a temptation to return to some sort of trajectory where when things are going moment to moment they're relatively predictable they don't require a relatively large amount of attention um and if you're going to enter into some uncertain uncertain change state i think you there's going to be a sort of collective pressure of a lot of brains regulating themselves to get back onto some predictable trajectory right you can't sustain like just complete uncertainty for a really long time because i don't think our brains can handle it so i would have to ask uncertainty about what uncertainty about the immediate lived environment so what's going on around you what are you doing when i got up today like i made coffee i got on the computer i have a predictable set of sensory signals that i'm going to be running into i have yeah yeah like a daily routine these things i don't mean in a broad sense yeah yeah because let's see so what was it climate change pandemic so you know most of these are not actually things that make your day-to-day life that unpredictable that's exactly what i'm saying i'm saying that those long-term events those things that you might be worrying about those things you know that we're talking about being worried about climate change they don't actually impact your day-to-day life right now very much and so changing your day-to-day life substantially now to deal with them is going to be difficult that's all i'm saying well not necessarily because the reference trajectory for actually dealing with them is not necessarily that unpredictable either different from what you're already doing doesn't mean unpredictable it's true but the period where you shift what you're doing to something different is and so i guess what i'm saying is that a change is something that requires a difference in trajectory from where something was already gone yeah but i think the thing about so the thing about our trajectory-based formulation of allostasis is that well the whole point is that you know if you let's say you plan to go to the store you're driving on the road you make a right turn is the right turn unpredictable no it's not it's a predicted baked in part of the reference trajectory change does not have to be it's a predicted outcome and it's a predicted outcome and it's a predicted outcome and it's a !

I think that would be unpredictable and i think we should avoid conflating those two

can i can i bring this back yeah examples of things we talked about in the dot zero so daniel brought up a really interesting point when he talked about um glucose stacking and i automatically sort of brought me to the idea of uh people who are have diabetes and those that are uh i think it's hypoglycemic unawareness they're the most brittle diabetics um and essentially what that speaks to is they have a large blind spot to sort of the cumulative effects of what's going on because the pancreas isn't working correctly doing that kind of regulation piece that blind spot and the and the severe consequences of that point but again i don't want to beat this like like a dead animal but because you are blind to something implies that you cannot take up that perspective and then the consequences of that are more severe when you absolutely don't know how your body is going to respond to that donut or not right so again it's it's we talked about it in the in the in the context of jordan was really good in clarifying we can't think of this in economic terms but i think we can think of it in terms of blind spot terms if we have a blind spot to this how do we respond to it how do seven billion people respond to it if well what they do is they revert back to the to the now that's what they do about their blind spot right because they just don't want to take up the perspective swap because it's way too challenging they might also sense there might be you might think of it also so right of this control hierarchy of that um if you're missing uh an ability to sort of sense and regulate some parameters in the hierarchy there might be downstream consequences of of glucose use that can be sensed but the problem is it's going to mean that you don't have as tight control you're basically like you're saying you have a blind spot in that hierarchy that isn't as tightly regulated and so it could potentially wander out of some of the bounds that are going to be uh adaptive or safe because you're performing some sort of second order control on it instead of um controlling it more tightly with some of the sensors that you would normally use right yeah but what do you think if i could just give one quick response actually on this idea um the leverage point of the control system is the true control variable so beyond being a good regulator having a volume knob in your head if there's a volume knob in the generative process the ultimate control situation is to just have the actual control of the volume knob and there's cases where that's possible there's case where it's not but where it is possible it's the leverage point and then jordan you just pointed to if you're missing this ability to either um sense x like we can't directly perceive our glucagon circulating level or responsivity across different tissues it's just too multifaceted and subliminal of an interoception directly different hormone balances and so on the technology and measurements are increasingly allowing that to be laser focused um without the ability direct to directly control those subliminal factors to either sense or act on them we can only apply upstream or downstream control so we could modify our niche or use signposting or stigmargy so that our environment dissuades us or make something impossible like time boxing or just putting something away um or downstream consequences like if like um having um uh some kind of supplement that helps with lactose digestion so we can't regulate most you can't put oxygen in the blood simply you can't simply change the amount of cholesterol and so actually being able to have a framework such as you're

describing where we can talk about the true physiological variable the actual substance of the essential variable but also recognize that we're only going to have interoceptive access into potentially noisy or variable proxies that are influenced by many things and they're going to be like sometimes multiple steps up or downstream like lightheadedness in the context of glucose regulation but that can happen for a lot of reasons so it can't be simply the cue so really interesting there um eli or anyone else

uh so i think we have to differentiate between how could you build an active inference control system in general and how are nervous systems built in specific how our nervous system is built in specific is an empirical question and i have pretty strong doubts that it's all just a matter of leverage

for pretty much the same reason that i have my doubts about it's all about avoiding change like there's just such wide variability of behavior in real life there are people who embrace change there are people who challenge themselves you know there are people who give up some of their leverage or control over some things and if we don't look at those if we sweep those actual behavioral instances under the rug of a sort of general principle that says well everyone ought to be very controlling and conservative all the time then i can't help but think it's a sort of scientific sin so so you like

dean and i were talking when you were out too about um because sorry about that by the way um normally she's a bit more allostatically resilient but she might be ill oh i hope she's feeling better but we were talking about because i think maybe maybe there's a misunderstanding because i don't think that we're saying that people only want to control or make their environments perfectly predictable right i think um part of what we were talking about right we were talking about this idea of shifting perspectives or this idea of um how do uh

uh so we were coming at it from the perspective of what a collaboration between psychology and engineering really gets you in the first place even right and what we were talking about is that part of what it does if people have the goal of trying to understand each other at the end of it and reach some sort of mutual predictability right so that they've developed some understanding that they can build a forward model from that they can actually uh they have a model of um of the environment that incorporates some understanding from the the other collaborator right there's going to be a period of that where you don't necessarily understand each other super well and there's going to be a long period of speaking to each other and not totally making sense and having to ask a lot of questions and having to search the space and what we are talking about is that from the perspective of fully exploring a space to actually develop a complete model of it or at least a better model of it there's actually some advantage of not having things um quickly achieve that predictability for each other if you quickly just gravitate you know if i answer a question and the other person just like immediately understood what i meant by it it might be because you know i answered it in a way that just happened to nail it for that one person but no one else in an audience is going to be able to understand it no one else who's listening in is going to be able to understand it and so by having some sort of imperfect process of gravitating toward making things predictable for each other you know we actually achieve some more explanation exploration and we sort of get a better sense of the space that we're actually moving through and we might generate a better model than we would if we just immediately converged right

yeah of course

but i think yes oh sorry i'll go ahead

you raised the difference between these two questions how is the nervous system built in specific or in particulars and how can you build or what is an active inference model in general so maybe return to the top question um sure yeah so i think this is one of those things that can get a little conflated when we have these collaborations of psychologists and engineers where from the engineering side if you're building something for uh so to speak the other acronym that you can form out of active inference you know ai instead of actin then you don't want something that works like a human nervous system necessarily well you may want to share you know a few properties but actually you're building it to solve some sort of task or to exhibit some sort of relatively well specified behavioral routine and

therefore it's going to be in most respects very completely different from a human being and that's actually not just because you're trying to avoid um what is it rossom's universal robots right where of course those were actually made of flesh and blood like it's not because you don't want your computer to grow feelings and decide to go on strike it's because a priori there's no reason to build a machine that works like a person in the first place maybe it would decide to go on strike but you know the space of things that you can build is just so much larger

that what you would usually do as an engineer is just aim towards the point of the space that you actually need for your task and that also implies a sort of converse proposition which is that the human nervous system is a pretty specific point in a space of things that could have evolved or even in the space of things that did evolve you know human being versus

what is it it's not a sponge it's that sea organism mentioned in the one of those first in papers that you know when it goes through a certain life transition it latches itself down onto a rock and goes through a little stage of metamorphosis where it eats its nervous system yeah c squirt thank you ian ian like those both evolved there is some space of possibilities that includes both the c squirt and the human being this implies that it's a pretty broad space of possibilities and if we're using active inference to model them both this implies a very very large space of possible active inference models formalisms controllers

so um eli uh if i've heard you correctly there you're sort of saying that artificial intelligence or tools that humans make um that might be modeled may or may not be modeled using active inference um they you know there's usually a an outcome or a goal that's that that we that's quite clear why we build them so it might be i don't know i want to build a coffee maker to make me coffee um and i might use some you know very intelligent um methods to to build that coffee machine but the out the outcome is clear it seems to me that with the nervous system we're still not clear on what the outcome is

so if it's you know if it if the fep is right then it's just to minimize free energy minimize surprise but am i right you're sort of questioning whether that is the that is the cup of coffee we're after at the end

um yeah and so we always have to sort of remember that dog walker just got here

um we always sort of have to remember that surprise means under a certain generative metal

[illegible]

we, because we're still allostatically and homeostatically existing. But to your point, I think we have to be clear that those two things are quite different. One is alive and the other is not. And both require math, but not the same math. And even among, I would actually caution, even among living systems, living involves metamorphosis. It involves death at a certain point. It involves reproduction. It doesn't just involve individual survival. That is not how an ecosystem works.

No, for sure. And I think what I was just going to add was that I think from an autopoietic perspective, I don't think anyone would disagree with that. I think that what someone would say is that you need to think about from a perspective of reproducing the relationships that create an organism, reproduction in that process, right, could be a process of recreating the organism or recreating the structures that create that organism when there's components of it that couldn't be repaired without reconstructing the entire thing, basically, right? So you have, you'll have death, you'll have any one of these systems are going to have components of it that can't be repaired. And if parts of it can't be repaired, then a way of basically ensuring that that system continues to exist in the relationships that compose it is to reproduce and create a new system that has those relationships, but has the parts that are going to terminally fall into disrepair over a certain period of time, start fresh so it can begin that cycle again, right?

Yeah. So this, that's actually an empirical claim about the evolution of aging, which is that aging came first, you know, at say the single cell level and then reproduction follows to cope with that. And I don't know enough about the history of life to know whether that's true or not.

If I could connect this very interesting theme to Dean's bridges and ramps, which are two different engineering projects and have two different sets of structures to modify the niche with. And also Ian's original point about the epigenome and the exposome and taking ecological developmental and evolutionary perspectives on human health. So the process of epigenetic canalization and the way in which epigenetic variability can become entrenched through genomic changes, like one protein

that has a bifunctional aspect over evolutionary time, when that region undergoes some chance duplication that could allow those two, um, tandem proteins to diverge and functions and like sub specialize, because there's a new capacity in the genome for the coding sequence to take on some of the

responsibilities that had been taken by a protein modification or some other dual functionalization of the ancestral protein. So the bridges are on the continuum towards modifying the environment, modifying the anatomy to embody the constraints of the environment, like the buoyancy of a fish.

And then behavior takes us closer and closer to the brain. And then behavior takes us closer and closer to that person jumping off the plane that Ian showed.

Whether it's like one footstep, which is many times a challenge for all kinds of various regions.

I mean, it can be really anything. I mean, it can be really anything. And that's kind of the challenge of, of movement.

To the, um, social risk settings and people having different profiles or envelopes of their risk in different areas

of being alone, of being with other people, all these different settings.

Um, but those are more ramp-like because in the end, especially when it's socially extended, you can't have the actual bodily, you can't embody the content of social variability, but your, your physical body can embody the content of being regularized with the abiotic niche.

But the brain can only be prepared in a way that, you know, is different from the body.

Hope that kind of brought it back to that epigenetics angle because behavior is even more epi than epigenetics.

And so allostatic are changes that are faster than it. So it makes sense to put it within a broader evolutionary perspective on how our priors got honed in to even this modality.

Yeah.

Yeah.

Well, in our last, um, 25 minutes or so, where do we go from dot to either the researchers or Ian, wherever you feel.

Um, the point you made Daniel about evolutionary, I, you know, I didn't mention evolutionary biology specifically at the beginning, but, um,

for me, that seems like a, um, maybe a blind spot here.

Um,

you know, I'm thinking also about Markov blankets and if you, you know, gut, gut microbiome, um, influencing my,

the, you know, the, um, neurotransmitters that I produce.

And do I, do I include my gut microbiome in, within my Markov blankets?

And if so, you know, what am I feeding my gut microbiome?

Do I include the food that I'm selecting within that Markov blanket?

And if so, do I include the fields that the food is growing in?

Um, cause that will affect the foods.

And if so, do I include the microorganisms in the, in the soil?

Cause that affects the, how the food, and then that will affect the microbes in my gut.

And that will affect me.

And then do I look at the weather?

Uh, so, you know, it's, it's more of a kind of communion, communion of subjects rather than, um,

bound objects,

but we have to start somewhere.

So as a bit of a jumping off from that,

I think with all active inference work, there's always this incredibly important question, which is we can use the framework across so many different scales.

And then, you know, I just had, um, Maxwell Ramstead tell me that they derived a theorem that literally every physical system has a Markov blanket.

And then I asked him, well, how do you pick out the ones that we're interested in studying?

And he said, sort of,

you know, basically he said, well, there isn't necessarily a theorem directing you to the most interesting, you know, or a metric that tells you which ones are interesting.

So that's still in the realm of just an investigator's choice.

And I think, you know, there, we really have to figure out how to ask,

when do we think we're going to get something interesting out of applying this framework to a given object of study or to a specific object of study?

And when do we think we'll be doing something sort of like re-deriving evolution?

We're like, it's useful to do as a formal exercise.

But to my knowledge, we haven't learned anything new about evolution from the free energy principle or active inference that we didn't know before,

mainly because people had already described evolution in terms of inference, you know, prior to describing it in terms of active inference.

I think another thing that's useful to think about, and I've been thinking about this a lot lately,

is in terms of what are some of the metaphors that we're using to scaffold the scientific questions that we're asking?

Because I think one of the things that's a big strength of this paper of Eli's, right,

is that the title itself is actually a sequence of two metaphors, which is interoception is modeling allostasis as control.

And so I think there's a sense that if we have particular questions, you know, if we set ourselves up to think what is the,

what's the sort of framework that we're using for thinking about the questions that are interesting or not, as Eli is saying, what are the systems that are interesting?

What sort of metaphor is a Markov blanket?

What sort of metaphor is active inference?

What are the sort of things that these are directing us to look at?

And does thinking of control in the way that this paper has been doing direct us in a different direction to think about things in like a slightly different way, maybe?

Can I tag on that right away?

Can I get your response to this, Jordan?

Can I tag on that right away?

sometimes as being continuous and progressive but are we talking whether we're talking markov blanket

or whether we're talking anything that's gap respecting is not necessarily gap filling

that's my question to you jordan what do you think i think it's i think it's more about the

gap respecting than the constantly searching for something to jab and wedge in it because that that

space that that eli spoke to that that possibilities thing that really matters in terms of staying alive

right it's not when you get out of the particulars max ramstead says there is no rule for that yet

that's that's what you have to self-organize so what do you think i don't know i don't know how

to answer that i'm trying to familiarize feel but but what i but i guess um so then jordan i think

you should do you want to prep what you're saying for a moment and i'll give an answer to that for sure or you just take over

so i think the thing is that at least in our lab we are a physiologically focused affective science lab right and so we sort of come to active inference with a set of systems of interest and a set of phenomena of interest in mind already and that doesn't necessarily help other people pick out what they want but it does indicate that you can bring what you want to i think active inference ends up being a bit of a potluck field at a certain point that's a good answer yeah yep one thought on that is the structure of a lab with a system or a question in mind is an epistemic framework that allows the investigators to actually take on bold new theoretical choices and explore different frameworks that are investigating a specified system of interest like we're studying blood pressure regulation in the context of professional dodgeball players or interoceptive awareness in intramural dodgeball players and then the lab exists to funnel that into different kinds of analyses and scenarios that help address that system like all of the labs studying different human diseases and sometimes with challenges and successes

making their work in the lab connect back to the real world and being able to take on new theoretical perspectives some which might not initially appear to be helpful until years of basic research have occurred around some molecule or gene family so yeah that's very interesting what maxwell added and i think speaks a lot to

many of the notions that we've been discussing like pushing at not having two realms in the metabasean with a highly rule-based and then leaving the outer level with where there is no strategy we just accept the priors without having metacognition on that level

because that's just the limits of how far the model was extended out yep

um do either of you authors want to add anything about dodgeball

oh yes yes i was going to tell i was going to tell everyone why it was dodgeball so um essentially in basketball you have to shoot to the hoop so you throw the ball goes it either goes into the hoop or it misses if you're passing you know the person either catches or you missed you know the ball is going to be able to do it in dodgeball so you can see the ball is coming at you and so the idea here is that you're picturing yourself trying to dodge and this lets us sort of merge the intuitions about predictive processing and control where the reference signal is to avoid contact with the ball

so therefore the stimulus becomes both a prediction error and an actual control error

so in life you're always playing dodgeball with something

i mean if you walk out into the middle of a traffic yes like

there's always some obscure possibility of being you know hit by a rock out of nowhere but hopefully most of the time in life you're not actually playing dodgeball with something well there's always something that can disrupt you right yeah there's always you have a system it can be disrupted in many ways i'm just kicking on it because i think again it's it's about metaphors right and i think when you're talking about basketball you're talking about having a goal you're talking about having um some sort of

externally defined goal that you're trying to perform actions to satisfy and then you tally up the score in some way of it right versus in dodgeball you can get hit and you're out and you know the same thing is true if your physiological systems can get disrupted and you're out you're dead right yeah but i guess my intuition is that in dodgeball the ball is coming for you in a way that most of the time it isn't physiologically no that's true metaphors aren't perfect but i'm just saying that yeah physiology is not adversarial no no no but there's one with the oh no sorry go ahead oh it just remind me each sport provides kind of a different metaphor none of them total or of course exact but there's games where you're avoiding the ball there's games where you have to hit an inanimate object and then it bounces off like wall ball um there's games with implements without implements there's uh marathoning there's sprinting there's throwing things then there's just one two three eleven all numbers of different players on ice as dean knows and each of these settings have their interoceptive challenges and social included in that model dean yeah i was just going to say the metaphor you can have you can have a lattice of rope that makes you a bridge and you can have a lattice of rope that makes you a fishing net one captures one allows you to cross one is in between one is enveloping and i think that's why we use metaphors but the point that i would make is that when you mesh something you still have those gaps and i think it's important to see the non-continuity in this i think that's where the math the theoretical the ability to bring it down to that level if it's not if it's not only one it's not perceived as only being one math if there are maths involved i think you create the conditions around which better understanding can move forward now to jordan's point it takes longer if you don't have the proper gauge of mesh to catch the smaller fish and that's i think what maxwell was speaking to it's up to you to decide whether you're catching guppies or tuna there's your there's your metaphor for the day guy

one short thought is the a zero to one which is a boundary we can talk about in the future is um when the entire or large subsets of the active inference ontology and frankly just the use of statistical modeling these kinds of transformations are not 2022 sinesh et al inventions with respect to control processes um once one undertakes the statistics toolkit they enable new visibilities into what was already latent every time somebody had shown something in terms of a cdf they didn't always show this but this is also taking logs of variables and you know changing cdf and pdf some of these atomic units already exist and this paper does an incredible job of putting the setting of interoception and the function of allostasis into the intersecting legacies of control theory and cybernetic approaches as well as more recent developing threads many of which don't have a formal basis initially like ecological psychology and affective modeling and so bringing those two areas together is kind of that psychologist engineer conversation that was brought up and that's like the dodgeball game the game now are they throwing if they hit each other do they knock each other out or are they sending each other you know and as we explored initially it's not clear and the in that uncertainty in the beginning of the exchange it's not like the dodgeballs are particles they're more like waves what will jordan and eli your post paper directions be or how long are you going to be continuing or what directions here jordan i think you should tell them about our lab meeting next week

oh yeah we're we're covering um so i think one of the things is we've talked about uh controlling variables here we were interested in um or i'm interested at least in thinking about uh other control perspectives there's a perspective perceptual control theory um to think about whether the uh that basically the behavior is a means of um controlling inputs or controlling sensory inputs to a system uh but more specifically i'm interested in um thinking about how we can apply control systems thinking to neuroscience more generally so i think people are very used to taking an fmri image for example uh bold imaging and then treating it as an input output system where they deliver some sort of stimuli to the brain they'll observe some output which will be a bold response in some region and then they'll interpret that as causal that they deliver stimuli they see an effect in the brain and they've detected they've identified some cause-effect relationship and the problem is that people have been doing that for 20 30 years now you have a million different strands of these supposed cause-effect relationships and you don't have a whole lot to structure them and a different way to think about it is that the brain is um is a negative feedback controlled system that's trying to regulate some uh homeostatic variables within it or allostatic when we consider them in the context that we're doing and so if we deliver some sort of stimulation to the brain we're basically perturbing sensory input to it and then we're seeing some perceptual response or we're seeing some restabilization within the brain to bring it back to homeostasis within a region um and we're not really getting at causal relationships we're looking at control systems that are being perturbed so that's coming off of the chapter that was by henry yin called the crisis in neuroscience which there's also a really great um brain inspired podcast episode about that too which is worth checking out and you like you have some ideas too to follow up to think about um control and affects right yeah so um to start with that i would sort of make a small amendment to the notion of a negative feedback control system um jordan can you catch the podcast link yeah sorry just good i could okay oh no i've got it um and say you know maybe we should be thinking about nervous systems as mixed feedback control systems

so sometimes you need to stabilize a state of sensitivity to the environment rather than resilience against the environment and that would involve positive feedback and that's sort of where i get into thinking about affect is that you know if you say that there's sort of these two modes of control positive feedback stabilized control for states that are very sensitive to the environment and then negative feedback purely negative feedback really it's positive feedback in the service of negative feedback over the long run and then purely negative feedback states that are trying to be insensitive to the insensitive to the environment and you have failure and success for either of them and all of a sudden it sort of starts to look and you can structure the feedback loops in different ways it starts to look more like some of the core properties of affect that jump out in our data and sorry that's a very hand wavy explanation and it's because like we really don't have like i don't have this in the form of more than a few notes on my ipad but there's should i link the other brain inspired episode that got you yeah yeah yes like that that one's good enough that i included a quote from it in i think in my slides from here that we had

a lot of the rudolph sepulcher quote yes yeah yeah control is to engineering what philosophy is to the humanities

wow wow very interesting directions ian and dean what are your next research and education directions

um i'll go first i'm gonna listen to those podcasts is my next steps and um i've just really enjoyed this conversation and um what i think you're all all trying to achieve is amazing stuff and it's whether it's whether um studying humans is going to help us make better ai on robotics or whether attempting to make better you know engineer better machines is going to help us understand humans or whether you know we're going to as you said uh jordan bounce off each other and improve our understanding both ways i think it's um it's a good thing over to dean thanks ian thank you ian thank you eli thank you jordan always thank you daniel i think um i think the last two and i want to thank dave douglas although he's not on a tile today as well because he was there for the um dot one it's it's really rewarding you when you can actually have a conversation with people like i mean normally it's throw up the dodge balls and then if you had this conversation in that's in that space you know it would basically be yelled out of the gym so they could get on with the game and then there's these people that would actually want to figure out well what's really going on here you know besides the person that leaves and has to get the ice pack because they got hit in the head or whatever and i think the i i think the idea that a bunch of individuals can sit down and have one of these perspective swap sessions because they're all really good translators in their particular area of focus really it kind of changes the idea around what we've been doing and we're trying to do an active inference which is you know create create a bigger tent create a sense that people with different backgrounds don't have to feel strange because they don't well i don't know as much about active inference as max ram said so i'm going to spend 10 000 hours to have a conversation about this that's not what we're about what we're trying to do is pull a bunch of people into the room and let them translate together swap those perspectives actually go away as ian said i mean i'm going to go away now and figure out what why really why does this laser idea work not just sort of in in a hypothesis in my brain but really go away and think about that for a bit and i think if we can do that more um in a you know semi-formal way you still have to cover the content and you still need to get the contextual background but to get to this place where we are today jordan to your point it takes time but man the rewards and the payoffs it's just huge and again it's again daniel all the work that he puts into doing this it's just it's awesome so thank you thank you yeah thank you this is really fun yeah great conversation so really appreciate it also as always all right till the dot three anytime you or anyone else want to continue or present any other time always welcome so farewell thank you very much bye awesome thanks